SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1024	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Code of Conduct
*I **Dinesh V (192524309)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: **Dinesh V (192524309)**

Annexure A – Case Study Analysis

1. Understand the Problem Statement

1.1 Problem Explanation

Pharmaceutical manufacturing industries must maintain controlled environmental conditions to ensure the quality, safety, and effectiveness of medicines. Factors such as temperature, relative humidity, differential pressure, airborne particle concentration, airflow velocity, and door status must be continuously monitored in cleanrooms, isolators, production areas, and warehouses.
In many facilities, environmental monitoring is performed using manual records or independent monitoring systems. These approaches may cause delays in identifying abnormal conditions, difficulties in maintaining accurate records, and a lack of centralized monitoring across multiple facilities. If environmental deviations remain undetected, they may affect product quality, lead to batch rejection, and create regulatory compliance issues.

The proposed Pharmaceutical Manufacturing Environmental Monitoring Platform is a distributed software system that uses IoT sensors to collect environmental data continuously. It compares the collected readings with approved Good Manufacturing Practice (GMP) limits, identifies environmental excursions, sends alerts to responsible personnel, and helps assess the potential impact on pharmaceutical batches.

The system also maintains historical records, supports electronic batch record and Corrective and Preventive Action (CAPA) workflows, and generates audit-ready reports. Its primary purpose is to improve environmental monitoring, reduce response time, and support pharmaceutical quality management.

1.2 Objectives of the Proposed System

The main objectives are:

- To continuously collect environmental data from pharmaceutical manufacturing areas using IoT sensors.
- To monitor temperature, humidity, differential pressure, particle counts, airflow velocity, and door status.
- To compare environmental readings against approved product-specific and zone-specific GMP limits.
- To detect environmental excursions and abnormal trends in real time.
- To notify Quality Assurance (QA), operations, and facility management teams about deviations.
- To identify potentially affected pharmaceutical batches and support quality-impact assessments.
- To maintain traceable environmental records and support electronic batch record integration.
- To support investigations and CAPA workflows.
- To generate automated compliance, historical, and audit reports.
- To provide centralized monitoring across multiple pharmaceutical manufacturing facilities.
- To improve data integrity, system security, reliability, and regulatory inspection readiness.

1.3 Scope of the Problem

The scope of the proposed system includes the following areas:

- **Environmental monitoring:** Continuous collection and visualization of environmental parameters from cleanrooms, isolators, production zones, and warehouses.
- **Excursion detection:** Automatic comparison of sensor readings with approved environmental limits and identification of deviations.
- **Alert management:** Generation of notifications, alarm acknowledgement, escalation, and recording of response actions.
- **Batch-impact assessment:** Identification of batches that may have been exposed to environmental deviations, based on production timelines and zone information.

- **Quality management:** Support for deviation investigations, corrective actions, preventive actions, and QA review.
- **Data management:** Storage of historical readings, device information, approved limits, excursion records, and audit events.
- **Reporting:** Generation of environmental monitoring summaries, deviation reports, trend analyses, and audit evidence.
- **Multi-facility support:** Centralized monitoring of multiple sites with facility-specific configurations and access permissions.

The project focuses on environmental monitoring and decision support. Final decisions regarding product release, rejection, or disposition remain the responsibility of authorized quality personnel. Actual regulatory compliance also requires appropriate validation, procedures, calibrated instruments, and controlled operation.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders and Their Roles

1. Quality Assurance (QA) Team: Reviews environmental deviations, evaluates potential product impact, verifies investigation records, approves relevant quality decisions, and ensures that required documentation is available for inspections.

• **Challenges:** Delayed alerts, incomplete records, difficulty tracing affected batches, and time-consuming audit preparation.

2. Production and Operations Team: Operates manufacturing equipment and ensures that production activities occur under approved environmental conditions.

• **Challenges:** Unexpected environmental changes, production interruptions, and uncertainty about which batches may be affected.

3. Facility and Maintenance Team: Maintains HVAC systems, environmental monitoring equipment, sensors, and related infrastructure.

• **Challenges:** Identifying equipment faults, responding quickly to pressure or airflow problems, and maintaining sensor calibration schedules.

4. Laboratory and Environmental Monitoring Personnel: Monitors environmental conditions, reviews measurement results, and supports investigations into contamination risks.

• **Challenges:** Large volumes of readings, manual data entry, and difficulty identifying abnormal trends.

5. IT and System Administrators: Manages user accounts, access permissions, server infrastructure, backups, software updates, and cybersecurity.

• **Challenges:** Maintaining system availability, protecting sensitive records, managing multiple facilities, and recovering from network or server failures.

6. Regulatory and Compliance Auditors: Examines environmental records, deviation investigations, corrective actions, and evidence of compliance with applicable requirements.

• **Challenges:** Incomplete audit trails, inconsistent documentation, and difficulty retrieving historical records.

7. Pharmaceutical Manufacturing Management: Oversees manufacturing performance, operational costs, quality risks, and compliance across facilities.

• **Challenges:** Batch losses, production delays, limited visibility across sites, and increasing monitoring costs.

8. IoT Device and Sensor Providers: Supplies or maintains monitoring instruments, gateways, and communication interfaces.

• **Challenges:** Device compatibility, calibration accuracy, network reliability, and consistent data transmission.

9. External System Providers: Includes electronic batch record systems, laboratory information systems, and quality management systems that may exchange data with the platform.

• **Challenges:** Data-format differences, integration complexity, synchronization problems, and maintaining consistent records across systems.

10. Pharmaceutical Quality and Regulatory Management: Defines approved environmental requirements, oversees risk management, and ensures that the system is suitable for its intended regulated use.

• **Challenges:** Keeping configurations controlled, demonstrating system suitability, and maintaining documented evidence when requirements or software change.

2.2 Major Challenges Faced by Stakeholders

The major challenges identified across the enterprise include:

- **1. Delayed detection:** Environmental deviations may not be identified quickly when monitoring depends on manual checks.
- **2. Data management difficulties:** Large volumes of environmental readings are difficult to organize and analyze manually.
- **3. Batch quality risks:** It may be difficult to determine which products or batches were exposed to abnormal conditions.
- **4. Communication gaps:** QA, operations, and maintenance teams may not receive information at the same time.
- **5. Regulatory documentation:** Preparing complete and traceable records for inspections can require considerable effort.
- **6. Sensor and network failures:** Faulty devices or communication interruptions may cause missing or unreliable measurements.
- **7. Multi-site complexity:** Different facilities may have different monitoring requirements, thresholds, equipment, and access permissions.
- **8. Data security:** Unauthorized access or uncontrolled changes can compromise record integrity.
- **9. Integration problems:** Connecting environmental monitoring data with batch records and quality management systems can be technically complex.

- **10. Operational costs:** Batch rejection, investigations, equipment failures, and production interruptions can increase costs.

3. Analyze the Current Approach

3.1 Existing System or Current Process

In a traditional pharmaceutical environmental monitoring process, environmental conditions are monitored using standalone instruments, local monitoring applications, manual inspection records, or a combination of these methods.

Personnel periodically check environmental readings, record the measurements, and compare them with acceptable limits. If a deviation is identified, the responsible team is informed and an investigation is initiated. Batch records and environmental logs may then be reviewed manually to determine whether products were affected.

Some modern facilities already use automated monitoring systems. However, the level of automation, centralized visibility, integration, and reporting varies between facilities. Therefore, the proposed system aims to address the limitations that may remain in fragmented or insufficiently integrated monitoring environments.

3.2 Strengths of the Existing System

- **1. Simple implementation:** Paper-based logs and standalone instruments may be easy to introduce in small facilities.
- **2. Direct measurement:** Environmental instruments can provide readings from the monitored area.
- **3. Human supervision:** Trained personnel can interpret unusual conditions and investigate the surrounding circumstances.
- **4. Established procedures:** Existing monitoring and quality procedures may already be familiar to employees.
- **5. Low initial software requirements:** A manual system may not require a complex centralized software platform.

3.3 Limitations of the Existing System

- **1. Manual recording errors:** Incorrect entries, missing values, and inconsistent documentation may occur.
- **2. Delayed alerts:** Periodic checks may not identify deviations immediately.
- **3. Limited integration:** Independent systems may not share information with electronic batch records or quality management applications.
- **4. Difficult batch tracing:** Linking an excursion to all potentially affected batches can require manual investigation.
- **5. Time-consuming reporting:** Audit reports may require collecting data from different records and systems.

- **6. Limited trend analysis:** Historical measurements may not be analyzed effectively to identify recurring environmental problems.
- **7. Communication delays:** Information may not reach QA, production, and maintenance personnel simultaneously.
- **8. Limited multi-site visibility:** Management may lack a unified view of environmental conditions across facilities.
- **9. Device-failure visibility:** Missing readings may not be distinguished promptly from normal conditions.
- **10. Data-integrity challenges:** Inadequate controls may make it difficult to demonstrate who changed a record, when it was changed, and why.

3.4 Problems and Gaps That Need to Be Addressed

The major gaps in the existing approach are the lack of continuous centralized monitoring, automated excursion detection, timely notification, batch-impact traceability, integrated quality workflows, and efficient report generation.

The proposed system should also identify sensor failures, preserve historical information, and prevent unauthorized modifications to critical records.

The goal is not to remove human supervision. Instead, the software should provide accurate and timely information so that qualified personnel can investigate deviations and make appropriate quality decisions.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Proposed Software Solution

The proposed solution is a distributed Pharmaceutical Manufacturing Environmental Monitoring Platform that integrates IoT devices, a backend processing system, a database, and a web-based monitoring dashboard.

Sensors continuously collect environmental readings and transmit them securely to the platform. The backend validates incoming data, compares measurements with approved limits, identifies excursions, and generates alerts. When a deviation occurs, the system records the event and identifies potentially affected batches using the relevant zone and production timeline.

The QA team can review the event, document investigations, and coordinate corrective and preventive actions. The platform stores historical data and produces reports for internal quality review and regulatory inspections.

The architecture should support multiple facilities while maintaining facility-specific configurations, role-based permissions, and reliable monitoring during component or network failures.

4.2 Major Functional Modules

- **Module 1: Facility and Zone Management —** Registers facilities, buildings, cleanrooms, isolators, production areas, and warehouses. It associates sensors with monitoring zones and manages the relevant configurations.
- **Module 2: IoT Sensor Data Collection —** Receives temperature, humidity, differential pressure, particle counts, airflow velocity, and door-status readings. It validates incoming data and identifies missing, stale, or invalid measurements.
- **Module 3: Environmental Threshold Management —** Maintains approved product-specific and zone-specific environmental limits. It supports controlled configuration changes and records the applicable threshold version.
- **Module 4: Real-Time Monitoring and Excursion Detection —** Displays live readings and compares them against approved limits. It detects excursions, identifies abnormal trends, and records the relevant events.
- **Module 5: Alert and Notification Management —** Sends notifications to QA, operations, and facility teams. It records acknowledgement status and supports escalation when required actions are delayed.
- **Module 6: Batch-Impact Assessment —** Associates environmental excursions with production zones and batch time windows. It identifies potentially affected batches and presents relevant readings and timelines for QA investigation.
- **Module 7: Deviation and CAPA Management —** Maintains deviation records, investigation details, corrective actions, preventive actions, responsibilities, and closure status.
- **Module 8: Electronic Batch Record Integration —** Exchanges relevant environmental records and excursion information with electronic batch record systems through controlled interfaces.
- **Module 9: Reporting and Analytics —** Generates environmental summaries, historical trends, excursion reports, investigation reports, and audit-ready documentation.
- **Module 10: User, Security, and Audit Management —** Manages authentication, role-based permissions, audit trails, record-history controls, and electronic-signature workflows where required.

4.3 Functional Requirements

Functional requirements describe the specific operations the system must perform:

- The system shall register facilities, zones, sensors, and users.
- The system shall collect and store environmental telemetry continuously.
- The system shall display current readings and historical trends.
- The system shall compare readings against approved environmental limits.
- The system shall detect excursions and configured abnormal trends.
- The system shall notify the relevant personnel when deviations occur.
- The system shall record alert acknowledgement and escalation information.

- The system shall identify potentially affected batches based on zone and production time information.
- The system shall support deviation investigations and CAPA workflows.
- The system shall generate reports for defined time periods and monitoring zones.
- The system shall maintain traceable records of relevant user actions and changes.
- The system shall support controlled data exchange with electronic batch record systems.
- The system shall identify missing sensor readings and device communication failures.
- The system shall support centralized monitoring of multiple facilities.

4.4 Non-Functional Requirements

- **Reliability:** The system must handle equipment and network failures without silently losing critical monitoring information.
- **Performance:** It must process incoming readings and issue configured alerts within defined response-time requirements.
- **Scalability:** It must support additional facilities, monitoring zones, sensors, and users without major architectural redesign.
- **Security:** It must use secure communication, authentication, authorization, and appropriate protection for sensitive information.
- **Data integrity:** It must preserve accurate readings, timestamps, record history, and traceability of critical changes.
- **Availability:** It should provide continuous monitoring through fault detection, buffering, recovery mechanisms, and suitable redundancy.
- **Usability:** The dashboard must present environmental readings, alarms, trends, and affected batches clearly.
- **Maintainability:** The software should be modular, documented, testable, and easy to update through controlled changes.
- **Interoperability:** The platform should support secure and consistent data exchange with relevant external systems.
- **Compliance readiness:** It should support applicable electronic-record controls, audit trails, validation evidence, and electronic signatures where required.

4.5 Application of Software Engineering Principles

- **1. Modularity:** The system is divided into separate modules for sensor ingestion, threshold evaluation, alerts, batch-impact assessment, reporting, and user management. Each module has a defined responsibility, making development and testing easier.
- **2. Scalability:** A distributed architecture allows additional facilities and sensors to be added as the manufacturing network grows. Independent processing and storage components can be scaled when necessary.
- **3. Maintainability:** Clear module boundaries, source-code version control, automated tests, documentation, and controlled configuration changes make the software easier to maintain.

- **4. Reliability:** The system includes device-health monitoring, local data buffering, communication retries, backup mechanisms, and recovery procedures. Missing data must be explicitly identified rather than treated as normal environmental conditions.
- **5. Usability:** The user interface provides live dashboards, colour-coded alerts, historical graphs, search facilities, and role-specific views for QA, operations, and administrators.
- **6. Security:** Role-based access control, secure transmission, strong authentication, audit logging, and protection against unauthorized record modification help safeguard the platform.
- **7. Abstraction:** Sensor communication, business logic, database operations, and the user interface are separated so that implementation details remain hidden from unrelated components.
- **8. Reusability:** Common components such as notification services, authentication, report generation, and threshold evaluation can be reused across different facilities.
- **9. Testability:** Modules have defined inputs and outputs, allowing independent unit testing and automated verification of system behaviour.
- **10. Traceability:** Requirements, test cases, approved configurations, deviations, corrective actions, and software changes should be linked through documented records.

5. Justify the Chosen Methodology/Framework

5.1 Selected SDLC Model: V-Model with Iterative Prototyping

The V-Model is selected as the primary software development lifecycle model because the proposed platform handles pharmaceutical manufacturing information that may influence product-quality investigations and regulated records.

The V-Model emphasizes systematic verification and validation. Every major development stage has a corresponding testing activity, which helps demonstrate that the implemented system satisfies its documented requirements.

Iterative prototyping is also useful during early development. A prototype can demonstrate sensor readings, dashboards, alerts, and batch-impact timelines to stakeholders before the complete system is developed.

5.2 Stages of the Selected Methodology

- **Stage 1: Requirements Analysis —** Collect requirements from QA personnel, production operators, maintenance teams, administrators, and compliance stakeholders. Document environmental parameters, thresholds, response expectations, access roles, and reporting needs.
- **Stage 2: System and Architectural Design —** Define the system architecture, database schema, interfaces, data flows, security controls, and failure-handling mechanisms.
- **Stage 3: Detailed Design —** Specify the behaviour of individual modules, including telemetry validation, excursion detection, alert generation, batch association, and audit logging.

- **Stage 4: Implementation —** Develop the software modules using suitable programming languages, databases, communication protocols, and development tools.
- **Stage 5: Unit Testing —** Test individual modules such as threshold comparison, data validation, notification logic, and report generation.
- **Stage 6: Integration Testing —** Verify that sensors or simulators, backend services, database components, notification services, and external interfaces exchange information correctly.
- **Stage 7: System Testing —** Evaluate the complete platform under normal operating conditions, environmental excursions, sensor failures, network interruptions, and high data volumes.
- **Stage 8: Acceptance Testing and Validation —** Verify that the system satisfies approved user requirements and is suitable for its intended use. In a regulated deployment, appropriate validation evidence and documented approvals are necessary before operational use.
- **Stage 9: Deployment and Maintenance —** Deploy the application, train users, monitor performance, correct defects, and manage software changes through controlled procedures.

5.3 Why the V-Model Is Suitable

- **1. Strong testing focus:** Each development stage has a corresponding testing activity.
- **2. Requirements traceability:** Requirements can be linked to design components and test evidence.
- **3. Risk reduction:** Critical failures can be identified systematically before deployment.
- **4. Documentation:** The model encourages documented requirements, design decisions, and test results.
- **5. Quality assurance:** It supports verification of excursion detection, security, reliability, and data-integrity controls.
- **6. Regulated-system suitability:** Its structured approach supports validation planning and documented evidence, although the model alone does not establish regulatory compliance.

5.4 Support for Successful Development and Maintenance

The selected approach begins with clear requirements and a structured design. A working prototype is then used to validate the main workflows and collect stakeholder feedback.

Once requirements and designs are sufficiently established, implementation proceeds with corresponding unit, integration, system, and acceptance testing. Defects are documented and corrected before release.

After deployment, maintenance activities follow controlled change procedures. Updates to environmental thresholds, software components, or external interfaces are reviewed, tested, and documented to preserve system reliability and record traceability.

This approach balances the need for systematic verification with the practical benefit of obtaining early feedback through a prototype.

6. Discuss Expected Outcomes and Limitations

6.1 Expected Benefits

- **1. Faster Detection of Environmental Deviations:** Continuous monitoring and automated threshold evaluation help identify excursions earlier than periodic manual checks.
- **2. Improved Product Quality Protection:** Timely alerts and batch-impact assessment support faster investigation of potentially affected products.
- **3. Improved GMP Monitoring & Inspection Readiness:** Centralized, traceable records and automated reports help personnel retrieve relevant evidence for audits and inspections.
- **4. Reduced Manual Work:** Automated collection, notifications, and reporting reduce repetitive data-entry and documentation tasks.
- **5. Better Communication:** QA, production, and maintenance teams receive relevant information through a shared monitoring platform.
- **6. Improved Historical Analysis:** Trend charts and historical records help identify recurring environmental deviations and potential equipment problems.
- **7. Centralized Multi-Site Visibility:** Management can monitor multiple facilities while maintaining site-specific environmental limits and access permissions.
- **8. Improved Data Traceability:** Audit records and controlled workflows help establish who performed relevant actions and when they occurred.
- **9. Support for Continuous Improvement:** Historical analytics and CAPA records can help organizations investigate recurring problems and evaluate corrective measures.
- **10. Potential Cost Reduction:** Earlier detection and improved investigations may reduce avoidable batch losses, investigation effort, and production interruptions.

6.2 How the Proposed Solution Addresses Identified Challenges

The table below maps the existing operational challenges to the technical solutions delivered by the proposed platform:

Existing Challenge	Proposed Solution
Manual environmental records	Automated sensor-data collection
Delayed detection	Continuous monitoring and threshold evaluation
Slow communication	Real-time notifications and escalation
Difficulty tracing affected batches	Time-based batch-impact assessment
Manual report preparation	Automated report generation
Limited historical analysis	Historical data storage and trend charts
Fragmented site information	Centralized multi-facility dashboard
Missing sensor readings	Device-health and missing-data alerts
Uncontrolled record changes	Access controls and traceable audit records
Disconnected quality workflows	Electronic batch record and CAPA integration

6.3 Potential Risks and Implementation Challenges

- **1. Sensor Accuracy and Calibration:** Incorrect measurements can lead to missed or false alarms. Sensors must be appropriately selected, maintained, and calibrated, with calibration status considered during monitoring.
- **2. Network and Power Failures:** Communication interruptions may prevent immediate data transmission. Local buffering, reliable recovery procedures, and monitoring of missing data are required.
- **3. Data Volume and Storage:** Continuous readings from many sensors create large datasets. Efficient storage, indexing, retention policies, and scalable processing are necessary.
- **4. Cybersecurity Threats:** Unauthorized access, compromised devices, or attacks on communication channels may threaten system security. Appropriate authentication, encryption, patching, and access controls are needed.
- **5. Integration Complexity:** Connecting different electronic batch record and quality-management systems may require custom interfaces, data mapping, and careful synchronization.
- **6. Regulatory Validation:** The software must be evaluated for its intended use, with documented requirements, risk assessments, test evidence, controlled changes, and appropriate validation. Compliance cannot be assumed simply because an audit-trail feature exists.
- **7. Incorrect Threshold Configuration:** An incorrect environmental limit can produce false alarms or fail to identify a genuine deviation. Approved limits should be version-controlled and subject to appropriate review and authorization.
- **8. False Alarms and Alert Fatigue:** Excessive notifications may reduce the attention given to important alarms. Alert rules should be risk-based, tested, and supported by appropriate escalation and acknowledgement procedures.
- **9. Batch-Impact Assessment Limitations:** Temporal overlap between a batch and an environmental excursion indicates potential exposure, not definitive product damage. Final quality assessments require documented investigation by authorized personnel.
- **10. Implementation and Training Costs:** Even when the software uses open-source tools, industrial deployment requires suitable hardware, infrastructure, cybersecurity, integration, training, and validation resources.

6.4 Limitations of the Proposed System

- The accuracy of monitoring depends on the quality and calibration of the sensors.
- Real-time alerting depends on the availability of communication and notification services.
- Automated rules cannot replace expert judgment in complex product-quality investigations.
- Historical trends do not guarantee that future environmental conditions will remain within limits.
- Integration with existing industrial systems may require additional development and testing.
- A student prototype using simulated sensor data cannot demonstrate the full reliability of a production installation.

- Regulatory compliance depends on the complete system, its intended use, validation, procedures, training, and ongoing control—not only on software features.
- Advanced analytics may produce incorrect results if the input data is incomplete, inaccurate, or poorly configured.

6.5 Future Enhancements

Future versions could include machine-learning-based anomaly detection, predictive maintenance, advanced trend analysis, redundant communication channels, improved cross-site analytics, and broader integration with laboratory and manufacturing systems.

Such enhancements should supplement validated monitoring and approved environmental limits rather than replace required controls.

Conclusion

The Pharmaceutical Manufacturing Environmental Monitoring Platform provides a software-based approach to improving environmental monitoring in pharmaceutical manufacturing facilities. By combining IoT data collection, real-time excursion detection, automated alerts, batch-impact assessment, quality-management workflows, and centralized reporting, the proposed solution addresses important limitations associated with fragmented and manual monitoring processes.

The application of Software Engineering principles such as modularity, scalability, reliability, maintainability, usability, security, and traceability provides a structured foundation for development. The V-Model, supported by iterative prototyping, is an appropriate development approach because it emphasizes systematic testing, documented requirements, and validation.

The proposed system can improve monitoring efficiency, support faster quality investigations, and strengthen audit readiness. However, successful implementation requires accurate sensors, secure infrastructure, reliable communication, suitable integration, trained users, and appropriate validation.

Overall, the project demonstrates how Software Engineering principles can be applied to develop a reliable, maintainable, and scalable software solution for a critical pharmaceutical manufacturing problem.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25